9. CORRELATION AND REGRESSION ANALYSIS

Exercise 9.1

1. 0.575
2. 0.947
3. 0.996
4. 0.891
5. 0.225
6. –0.0735
7. 0.9
8. 0.224
9. 0.905
10. –0.37

Exercise 9.2

1.(a) Y=–0.66X+59.12; X=–0.234Y+40.892

(b) r=-0.394 (c) Y= 39.32

2. Y=0.6102X+66.12; X=0.556Y+74.62 Height of son is 166.19
3. Y=2.3X–35.67; weight of the student is 125.79 lb
4. Y=0.24X+1.04; X=1.33Y+1.34
5. Y=1.6X; estimated yield = 46.4 unit per acre
6. Y=0.942X+6.08; Estimated sales =34.34 (in crores of rupees)
7. Y=0.48X+67.72; X=0.91Y–41.35;Y=72.52
8. b_{yx}=1.33; Y=1.33X+3.35
9. Y=0.1565X+19.94; estimated expenditure on food and entertainment (Y) is 51.24
10. X=0.8Y–1 and estimated value of X is 5.4 when Y=8

    Y=0.8X+2.6 and estimated value of Y is 12.2 when X=12

11. $\bar{X}$ = 13; $\bar{Y}$ =17 and r = 0.6
12. $b_{xy} = -\frac{3}{2}; b_{yx} = -\frac{1}{2}; r = -0.866$

Exercise 9.3

1	2	3	4	5	6	7	8	9	10	11	12	13
(a)	(b)	(a)	(b)	(c)	(a)	(a)	(b)	(a)	(b)	(a)	(a)	(c)
14	15	16	17	18	19	20	21	22	23	24	25	
(a)	(b)	(a)	(a)	(a)	(a)	(a)	(b)	(b)	(b)	(a)	(d)	

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